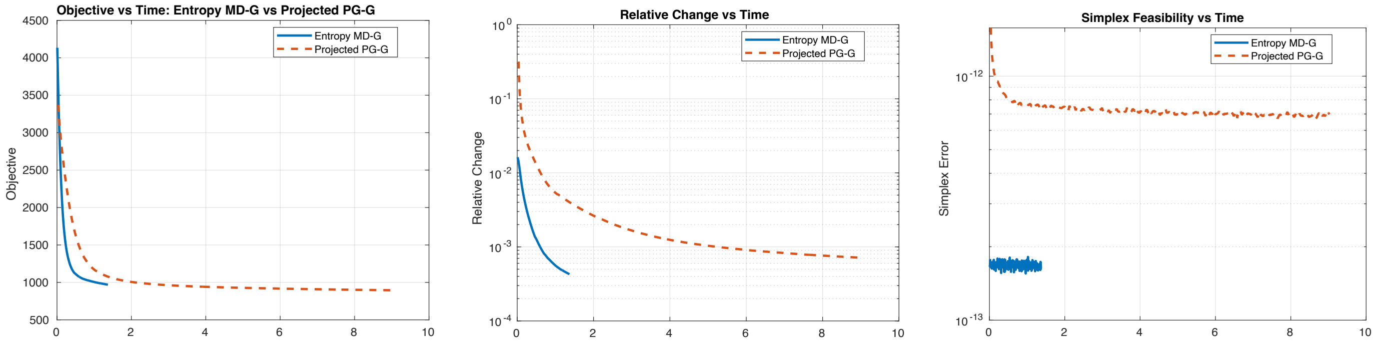

(a) Objective vs. time

(b) Relative change vs. time

(c) Simplex error vs. time

Fig. 2: Comparison between entropy mirror descent and Euclidean projected gradient for updating simplex-constrained matrix $\mathbf{G}$. Both methods use the same objective, initialization, and $\mathbf{F}$-update. The proposed **MD-**G update decreases the objective faster with respect to CPU time and maintains smaller simplex feasibility error on synthetic data $(n, d, k) = (5000, 100, 50)$.